

FACULTY OF SCIENCE
B.Sc. (VI-Semester) Examination
Botany

Paper – VI (b) (New)
Tissue Culture and Biotechnology

Time: 3 Hours)

Section – A (Marks: 8x4=32)

(Max. Marks: 80)

1. Attempt any **eight** questions in short-form.

- a) Explant
- b) Totipotency
- c) Anther culture
- d) Somaclonal Variations
- e) Hairy root cultures
- f) Synthetic seeds
- g) Applications of rDNA technology
- h) Cosmids
- i) Polymerases
- j) Genomic libraries
- k) Microprojectile gun
- l) Agroinfection

Section – B (Marks: 4x12=48)

Answer **all** questions

2. a) Explain the various sterilization techniques used in plant tissue culture.
(OR)
b) Describe the different types of organ cultures. Explain with suitable examples.
3. a) Explain somatic hybridization in plants and add a note on its applications.
(OR)
b) Describe cryopreservation and discuss its importance in germplasm conservation.
4. a) Explain the gene cloning process, including the steps and methods for selection of recombinant clones.
(OR)
b) Describe the types of restriction enzymes and ligases and their role in rDNA technology.
5. a) Explain the principle and steps of Polymerase Chain Reaction (PCR) and discuss its applications.
(OR)
b) Evaluate the application and advantages of transgenic technology in modern agriculture.

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FACULTY OF SCIENCE
B.Sc. (II Semester) Examination
BOTANY

Paper- II (New)
Gymnosperms, Anatomy and Embryology of Angiosperms

(Max. Marks: 80)

Time: 3 Hours)

Section - A (Marks: 8 x 4 = 32)

1. Attempt any **eight** questions
- a) Economic Importance of Gymnosperms ✓
 - b) Pinus male cone ✓
 - c) Importance of fossils ✓
 - d) Korper-Kappe theory ✓
 - e) Parenchyma tissue ✓
 - f) Fibers ✓
 - g) Heartwood and sapwood ✓
 - h) Types of vascular bundles ✓
 - i) Neem (*Azadirachta indica*) uses ✓
 - j) Nemec phenomenon ✓
 - k) Entry of the pollen tube into the ovule ✓
 - l) Triple fusion ✓

Section - B (Marks: 4 x 12 = 48)

Answer **all** the questions.

2. a) Explain about the structure of Male and Female strobili in *Gnetum*.
(OR)
b) Describe the anatomy of Pinus needle and its xerophytic characters.
3. a) Define meristematic tissue and write the various types of meristems and their functions.
(OR)
b) Explain the anatomy of Dicot leaf with a neat diagram.
4. a) Describe the anomalous secondary growth of *Boerhaavia* stem.
(OR)
b) Explain the Nallamaddi, Egisa and Neem wood structures and their uses.
5. a) Define the ovule? Write the various types of ovules.
(OR)
b) What is polyembryony? Describe various types of polyembryony.

Contd.